Bias and Variance, Overfitting, and the Support Vector Machine

5

Post

CS 3244
Machine Learning

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Recap and Forecast

Linear Regression – Real valued output

$$J(\theta) = \frac{1}{m}\sum_{j=1}^{m}\left(\theta^\top x^{(j)} - y^{(j)}\right)^2$$

Logistic Regression – Probability of an event

$$J(\theta) = \frac{1}{m}\sum_{j=1}^{m} ln(1 + e^{-y^{(j)}\theta^\top x^{(j)}})$$

Gradient Descent

- Take steps in opposite the gradient of the cost function.
- SGD: the average pointwise direction is equivalent to the gradient $-\nabla J$

Learning Outcomes for this week:

- Understand the bias–variance tradeoff
- Understand why overfitting occurs: the role of model complexity and the sampled dataset
- Apply bias and variance decomposition in simple scenarios

- Understand why maximizing the margin leads to an optimal separating hyperplane
- Understand why an SVM's model complexity is defined by its support vectors
- Extend SVMs to tolerate errors through soft margins

Bias and Variance

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Approximation–generalization tradeoff

Small J_{test}: good approximation of f on unseen test data

More complex $\mathcal{H}$ → better chance of **approximating** f

Less complex $\mathcal{H}$ → better chance of **generalizing** out of sample

Ideal $\mathcal{H} = \{f\}$ ← Good luck with that!

Quantifying the tradeoff

Decomposing J_{test} into

1. How well $\mathcal{H}$ can approximate f overall
2. How well we can zoom in on a good $h_\theta \in \mathcal{H}$

Applies to **real-valued** targets and uses **squared error**

There is an equivalent binary version through the lens of VC analyses (from SVM)

Start with J_{test}

$$J_{test}(h_{\mathcal{D}}) = \mathbb{E}_x[(h_{\mathcal{D}}(x) - f(x))^2]$$

$h(x)$ depends on the specific dataset $\mathcal{D}$.

Start with J_{test}

$$J_{test}(h_{\mathcal{D}}) = \mathbb{E}_x[(h_{\mathcal{D}}(x) - f(x))^2]$$

$h(x)$ depends on the specific dataset $\mathcal{D}$.

$$\mathbb{E}_{\mathcal{D}}[J_{test}(h_{\mathcal{D}})]$$

Generalizing over all $\mathcal{D}$ with same m

$$= \mathbb{E}_{\mathcal{D}}\left[\mathbb{E}_x[(h_{\mathcal{D}}(x) - f(x))^2]\right]$$

$$= \mathbb{E}_x\left[\mathbb{E}_{\mathcal{D}}[(h_{\mathcal{D}}(x) - f(x))^2]\right]$$

Swap ok, as integrand is strictly non-negative

Focus on $\mathbb{E}_{\mathcal{D}}[(h_{\mathcal{D}}(x) - f(x))^2]$, get the outer term $\mathbb{E}_x[\ldots]$ later.

The average hypothesis

To evaluate $\mathbb{E}_{\mathcal{D}}[(h_{\mathcal{D}}(x) - f(x))^2]$,
we define the 'average' hypothesis $\bar{h}(x) = \mathbb{E}_{\mathcal{D}}[h_{\mathcal{D}}(x)]$

Imagine many, many data sets $\mathcal{D}_1, \mathcal{D}_2, \ldots, \mathcal{D}_K$ drawn:

$$\bar{h}(x) \approx \frac{1}{K} \sum_{k=1}^{K} h_{\mathcal{D}}(x)$$

Using $\bar{h}(x)$

$$\mathbb{E}_{\mathcal{D}}[(h_{\mathcal{D}}(x) - f(x))^2] =$$

Using $\bar{h}(x)$

$$\mathbb{E}_{\mathcal{D}}[(h_{\mathcal{D}}(x) - f(x))^2] =$$

$$= \mathbb{E}_{\mathcal{D}}\left[\left(h_{\mathcal{D}}(x) - \bar{h}(x) + \bar{h}(x) - f(x)\right)^2\right]$$

Add $-\bar{h}(x) + \bar{h}(x)$.
Associate first two and second two terms.

$$= \mathbb{E}_{\mathcal{D}}\left[\left(h_{\mathcal{D}}(x) - \bar{h}(x)\right)^2 + \left(\bar{h}(x) - f(x)\right)^2 + 2\left(\underbrace{h_{\mathcal{D}}(x) - \bar{h}(x)}_{0}\right)\left(\underbrace{\bar{h}(x) - f(x)}_{\text{const w.r.t. } \mathcal{D}}\right)\right]$$

Expand out, cross terms drop as first part of term is 0.

$$= \mathbb{E}_{\mathcal{D}}\left[\left(h_{\mathcal{D}}(x) - \bar{h}(x)\right)^2\right] + \left(\bar{h}(x) - f(x)\right)^2$$

2nd term is constant with respect to $\mathcal{D}$

Bias and Variance

$$\mathbb{E}_{\mathcal{D}}[(h_{\mathcal{D}}(x) - f(x))^2] = \underbrace{\mathbb{E}_{\mathcal{D}}\left[\left(h_{\mathcal{D}}(x) - \bar{h}(x)\right)^2\right]}_{1} + \underbrace{\left(\bar{h}(x) - f(x)\right)^2}_{2}$$

Quick Quiz: Which is which?

Bias and Variance

$$\mathbb{E}_{\mathcal{D}}[(h_{\mathcal{D}}(x) - f(x))^2] = \underbrace{\mathbb{E}_{\mathcal{D}}\left[\left(h_{\mathcal{D}}(x) - \bar{h}(x)\right)^2\right]}_{\text{variance}} + \underbrace{\left(\bar{h}(x) - f(x)\right)^2}_{\text{bias}}$$

Therefore,

$$\mathbb{E}_{\mathcal{D}}[J_{test}(h_{\mathcal{D}})] = \mathbb{E}_x\left[\mathbb{E}_{\mathcal{D}}[(h_{\mathcal{D}}(x) - f(x))^2]\right]$$

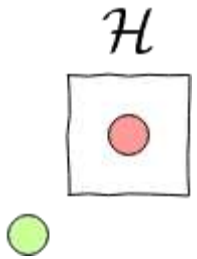
$$= \mathbb{E}_x[\text{bias}(x) + \text{var}(x)]$$

$$= \text{bias} + \text{var}$$

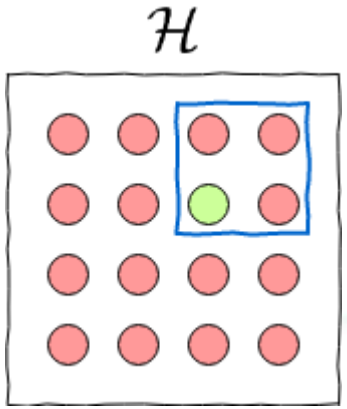
The tradeoff

$$\text{Bias}[h(x)] = \left|\bar{h}(x) - f(x)\right|$$

$$\text{Var}[h(x)] = \mathbb{E}\left[\left(h(x) - \bar{h}(x)\right)^2\right]$$



$\mathcal{H}_{small}$: just one hypothesis.



$\mathcal{H}_{big}$: big, contains the target function f. But you have to find it.

Overfit Measure

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An irony of two learners

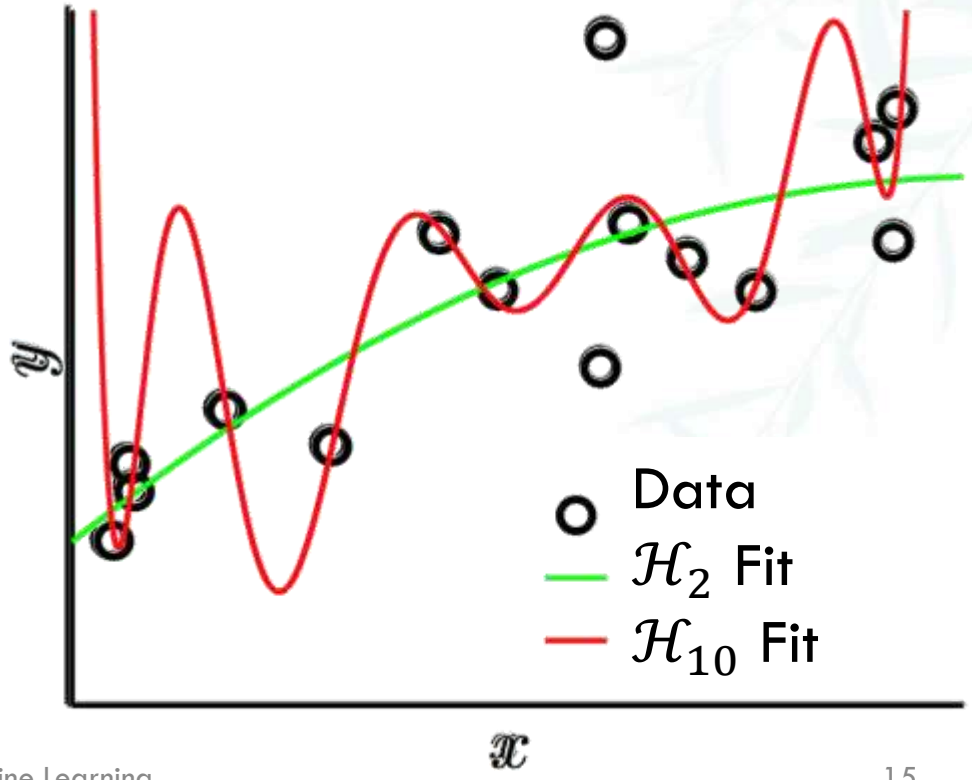
Two learners Overfitting and Restrained:

Both <u>know</u> the target is 10th order, you get 15 data points.

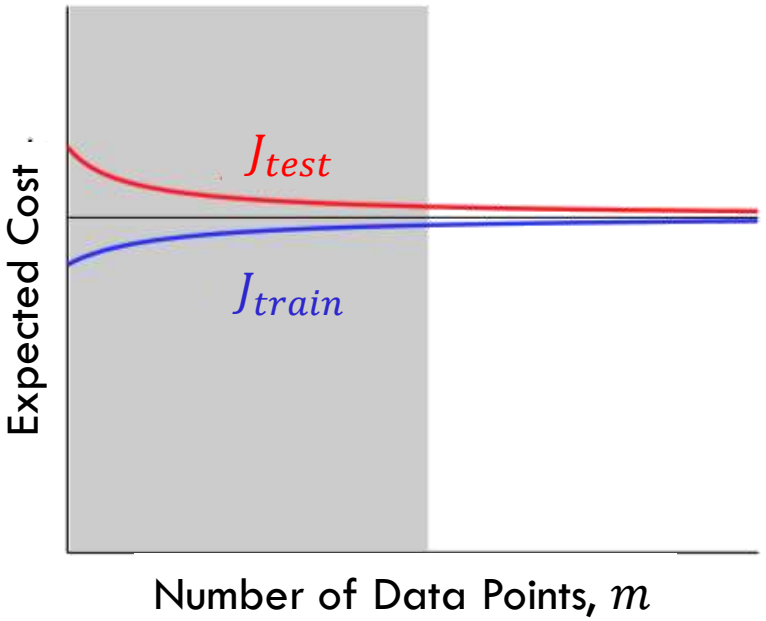
O chooses $\mathcal{H}_{10}$

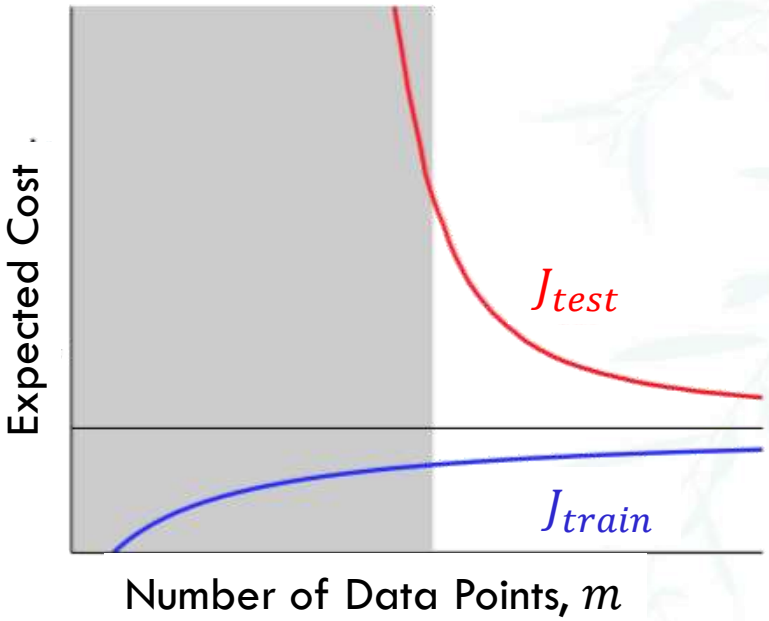
R chooses $\mathcal{H}_2$

Learning a 10th order target with noise



When is $\mathcal{H}_2$ better than $\mathcal{H}_{10}$?





Overfitting (grey region): $J_{test}(\mathcal{H}_{10}) > J_{test}(\mathcal{H}_2)$

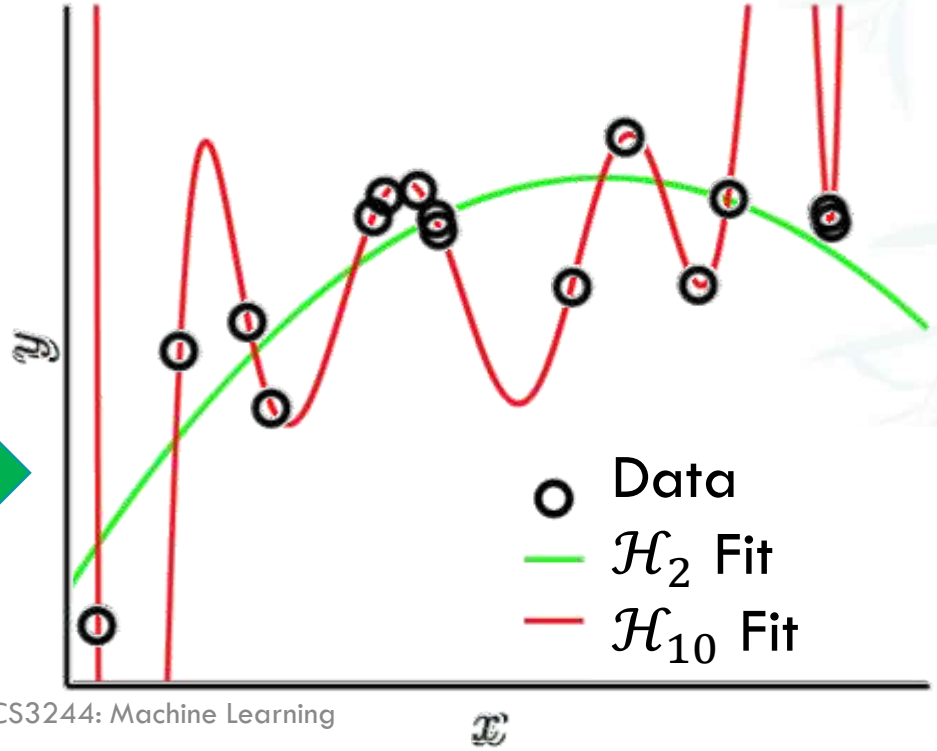
Even without noise with complex f

The two learners $\mathcal{H}_{10}$ and $\mathcal{H}_2$

They <u>know</u> there is no noise

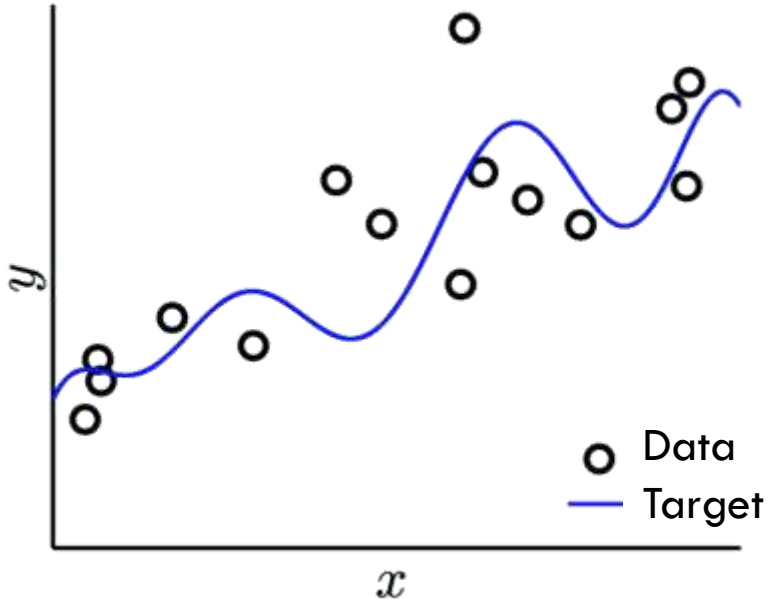
Is there really no noise?

Learning a 50th order target



A detailed experiment

Impact of **noise level** and **target complexity**



$$y = f(x) + \epsilon(x) = \underbrace{\sum_{q=0}^{Q_f} \alpha_q x^q}_{*} + \underbrace{\epsilon(x)}_{\sigma^2}$$

Noise level: σ^2

Target complexity: Q_f

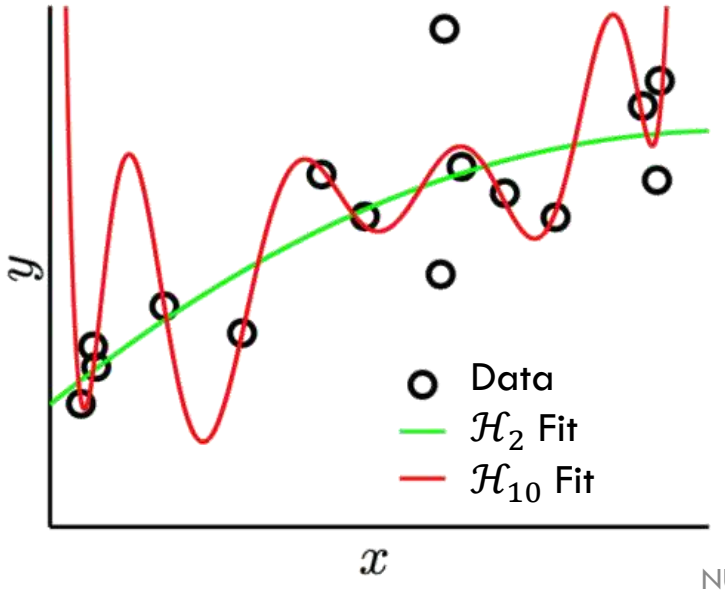
Data set size: m

The overfit measure

We fit the data set $(x_1, y_1), \ldots, (x_m, y_m)$ using our 2 models:

$\mathcal{H}_2$: 2nd order polynomials

$\mathcal{H}_{10}$:10th order polynomials

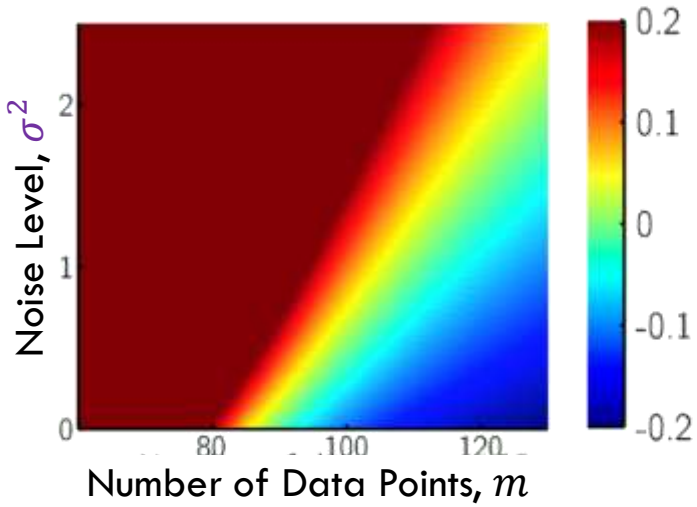


Compare out-of-sample errors of $h_2 \in \mathcal{H}_2$ and $h_{10} \in \mathcal{H}_{10}$

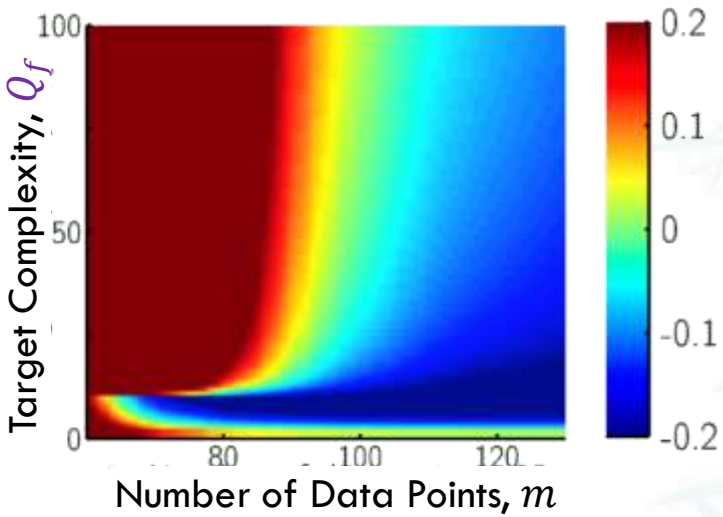
Overfit measure:

$$J_{test}(h_{10}) - J_{test}(h_2)$$

Overfit measure: $J_{test}(h_{10}) - J_{test}(h_2)$



Stochastic Noise



Deterministic Noise

Number of data points	↑	Overfitting ↓
Stochastic Noise	↑	Overfitting ↑
Deterministic Noise	↑	Overfitting ↑

The Role of Noise

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Noise

That part of y that we **cannot** model

It has two sources …

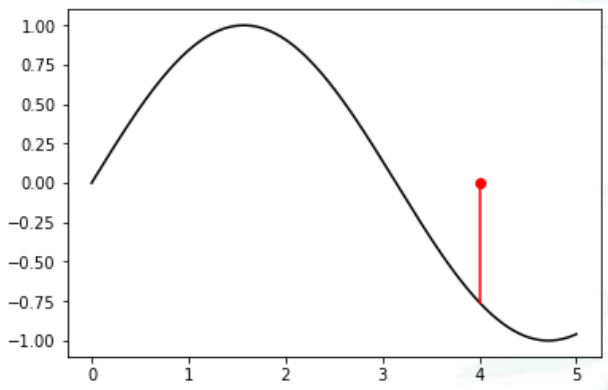
Stochastic Noise: Data Error

We would like to learn f

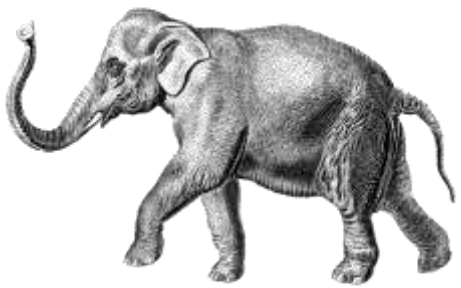
$$y_j = f(x_j)$$

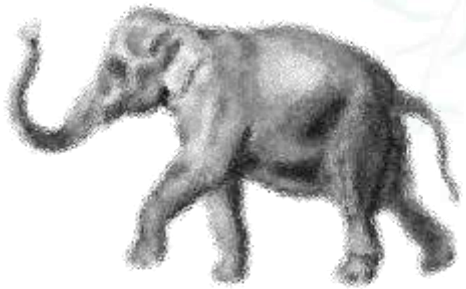
Unfortunately, we only observe $\mathcal{D}$

$$y_j = f(x_j) + \texttt{stochastic_noise}$$



Stochastic Noise
Fluctuations that
we cannot model





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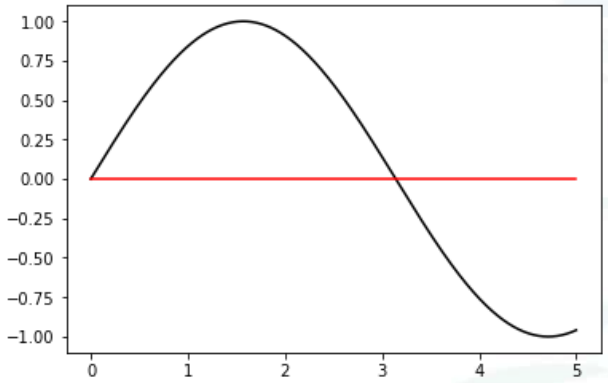
Deterministic Noise: Model Error

We would like to learn from O

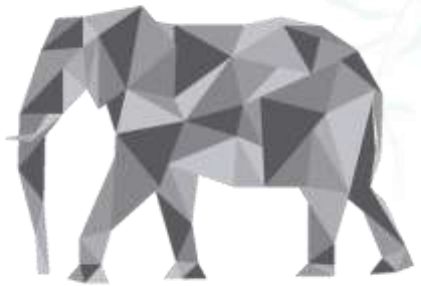
$$y_j = \bar{h}(x_j)$$

Unfortunately, we only observe $\mathcal{D}$

$$y_j = f(x_j)$$
$$= \bar{h}(x_j) + \texttt{deterministic_noise}$$



Deterministic Noise
The part of f we lack the capacity to model



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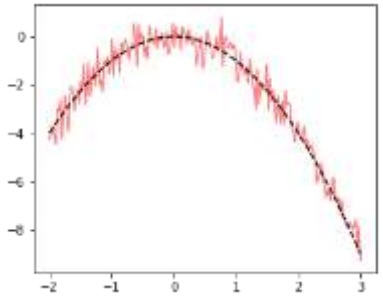
Both sources of noise hurt learning

Stochastic Noise

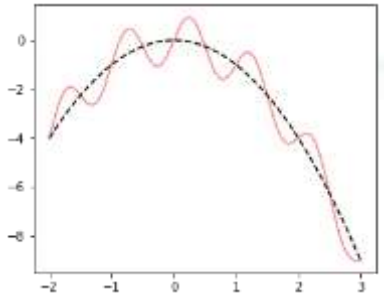


Source: random measurement errors

Re-measure y: Stochastic noise changes

Change $\mathcal{H}$: Stochastic noise still the same

Deterministic Noise



Source: learner's $\mathcal{H}$ cannot model f

Re-measure y: Deterministic noise the same

Change $\mathcal{H}$: Deterministic noise changes

We have a single $\mathcal{D}$ and fixed $\mathcal{H}$ so we cannot distinguish between either; with finite m, $\mathcal{H}$ will try to fit the noise.

Noise and Bias-Variance

Recall the decomposition:

$$\mathbb{E}_{\mathcal{D}}[(h_{\mathcal{D}}(x) - f(x))^2] = \underbrace{\mathbb{E}_{\mathcal{D}}\left[\left(h_{\mathcal{D}}(x) - \bar{h}(x)\right)^2\right]}_{\text{variance}} + \underbrace{\left(\bar{h}(x) - f(x)\right)^2}_{\text{bias}}$$

What if f is a noisy target?

$$y = f(x) + \epsilon(x) \qquad \mathbb{E}[\epsilon(x)] = 0$$

A noise term

$\mathbb{E}_{\mathcal{D},\epsilon}[(h_{\mathcal{D}}(x) - y)^2] =$

A noise term

$$\mathbb{E}_{\mathcal{D},\epsilon}[(h_{\mathcal{D}}(x) - y)^2] =$$

$$= \mathbb{E}_{\mathcal{D},\epsilon}[(h_{\mathcal{D}}(x) - f(x) - \epsilon(x))^2]$$

Expand observed y

$$= \mathbb{E}_{\mathcal{D},\epsilon}\left[\left(\underbrace{h_{\mathcal{D}}(x) - \bar{h}(x)}_{①} + \underbrace{\bar{h}(x) - f(x)}_{②} - \underbrace{\epsilon(x)}_{③}\right)^2\right]$$

Associate terms

$$= \mathbb{E}_{\mathcal{D},\epsilon}\left[\left(h_{\mathcal{D}}(x) - \bar{h}(x)\right)^2 + \left(\bar{h}(x) - f(x)\right)^2 + (\epsilon(x))^2 + \cancel{\text{cross terms}}\right]$$

Additional cross terms disappear as $\mathbb{E}(\epsilon(\boldsymbol{x})) = 0$

Actually two noise terms

$$\mathbb{E}_{\mathcal{D}}[(h_{\mathcal{D}}(x) - f(x))^2] =$$

$$\underbrace{\mathbb{E}_{\mathcal{D},x}\left[\left(h_{\mathcal{D}}(x) - \bar{h}(x)\right)^2\right]}_{\text{variance}} + \underbrace{\mathbb{E}_{x}\left(\bar{h}(x) - f(x)\right)^2}_{\text{bias}} + \underbrace{\mathbb{E}_{\epsilon,x}\left(\epsilon(x)\right)^2}_{\sigma^2}$$

Deterministic Noise

Stochastic Noise

Knn Regression with Noise

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k-nearest Regression with Noise

Assume $y^{(j)} = f(x^{(j)}) + \epsilon$ where $\mathbb{E}(\epsilon) = 0$ & $Var(\epsilon) = \sigma^2$.

Under squared error, the expected prediction error of a regression fit $f(x)$ at an input point $x = x^{(0)}$ can be written as the sum of Stochastic error, Bias^2 and Variance.

For the k-nearest regression fit, the error can be expressed as:

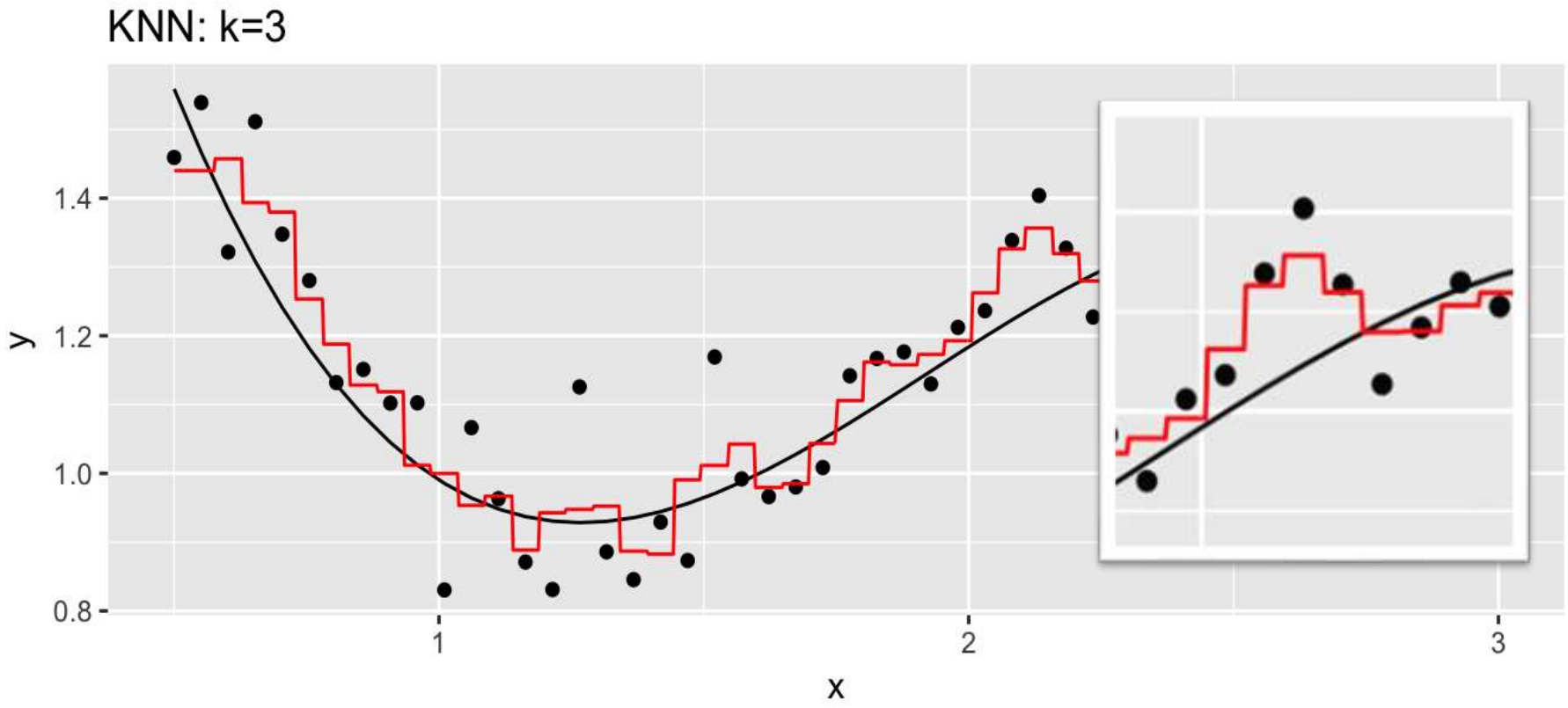
$$J(x^{(0)}) = \mathbb{E}[\left(y^{(0)} - h\left(x^{(0)}\right)\right)^2 |x^{(0)})]$$

$$= \sigma^2 + (f\left(x^{(0)}\right) - \frac{1}{k}\sum_{j=1}^{k} f(x^{(j)}))^2 + \frac{\sigma^2}{k}$$

where $x^{(j)}$ $(j = 1, \dots, k)$ are the k nearest data points.

Derive the above components of the equation.

kNN Regression in action



Derivation

- Stochastic Error σ^2
- Bias^2 $\left(f\left(x^{(0)}\right) - \frac{1}{k}\sum_{j=1}^{k} f\left(x^{(j)}\right)^2\right.$
- Variance $\frac{\sigma^2}{k}$

Bias^2

Bias^2

$$\left(f\left(x^{(0)}\right) - \mathbb{E}_{\mathcal{D}}\left[h_k\left(x^{(0)}\right)\right]\right)^2 = \left(f\left(x^{(0)}\right) - \mathbb{E}_{\mathcal{D}}\left[\frac{1}{k}\sum_{j=1}^{k} f\left(x^{(j)}\right)\right]\right)^2$$

$$= \left(f\left(x^{(0)}\right) - \frac{1}{k}\sum_{j=1}^{k} f\left(x^{(j)}\right)\right)^2$$

$$= \left(f\left(x^{(0)}\right) + \epsilon - \frac{1}{k}\sum_{j=1}^{k} f\left(x^{(j)}\right) + \epsilon\right)^2$$

$$= \left(f\left(x^{(0)}\right) - \frac{1}{k}\sum_{j=1}^{k} f\left(x^{(j)}\right)\right)^2$$

Assuming fixed neighbors $x^{(j)}$,

$$\mathbb{E}_{\mathcal{D}}\left[\frac{1}{k}\sum_{j=1}^{k} f\left(x^{(j)}\right)\right] = \frac{1}{k}\sum_{j=1}^{k} f\left(x^{(j)}\right)$$

Over all possible $\mathcal{D}$, ϵ values disappear as they have zero mean

Variance

Variance

$$
\begin{aligned}
Var(h_k(x^{(0)}) &= Var(\frac{1}{k}\sum_{j=1}^{k} f(x^{(j)})) \\
&= \frac{1}{k^2}\sum_{j=1}^{k} Var(f(x^{(j)}) + \epsilon^{(j)}) \\
&= \frac{1}{k^2}\sum_{j=1}^{k} Var(f(x^{(j)})) + Var(\epsilon^{(j)}) \\
&= \frac{1}{k^2}\sum_{j=1}^{k} Var(\epsilon^{(j)}) \\
&= \frac{1}{k^2}\sum_{j=1}^{k} \sigma^2 \quad = \frac{\sigma^2}{k}
\end{aligned}
$$

$Var(aX) = a^2 Var(X)$, as $\mathbb{E}(aX) = a\mathbb{E}(X)$

$Var(f(x^{(j)})) = 0$, as we assume the neighbors $x^{(j)}$s are fixed, and hence have no variance.

Learning Curves

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Learning Curves

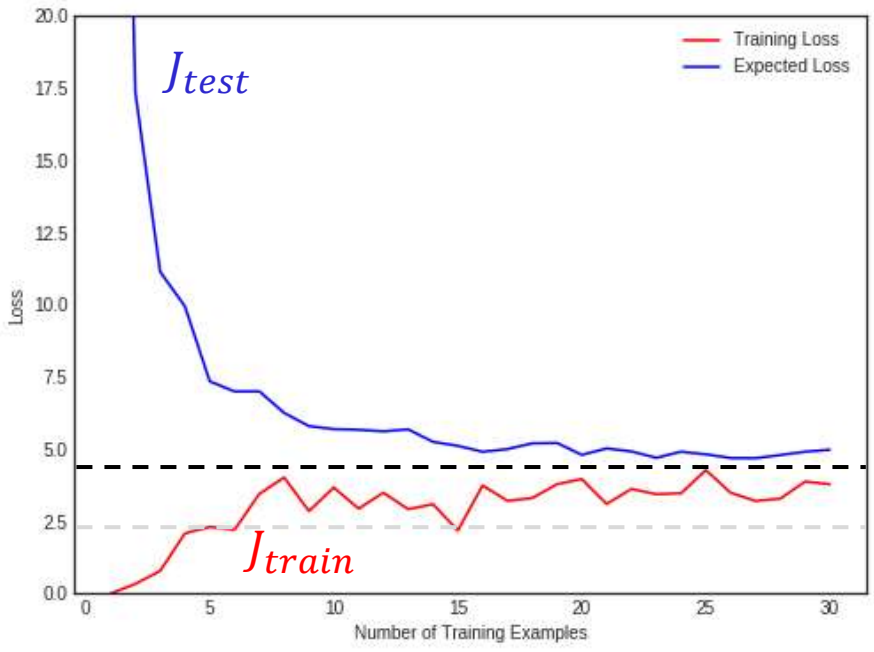
Plot expected J_{test} and J_{train} …

…as we vary size m

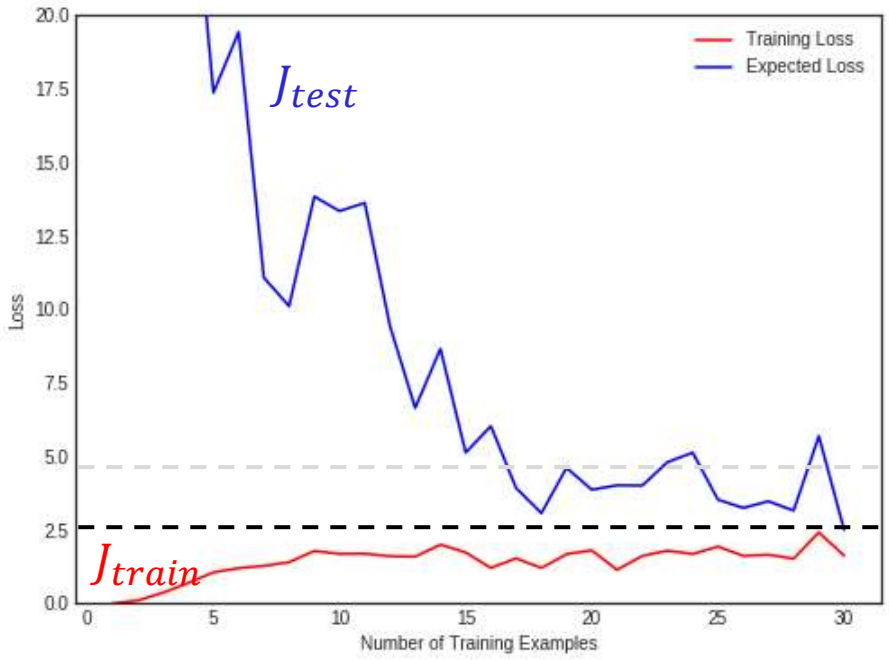
Expected test cost $\mathbb{E}_{\mathcal{D}}[J_{test}(f_{\mathcal{D}})]$

Expected training cost $\mathbb{E}_{\mathcal{D}}[J_{train}(f_{\mathcal{D}})]$

Simple versus Complex Models

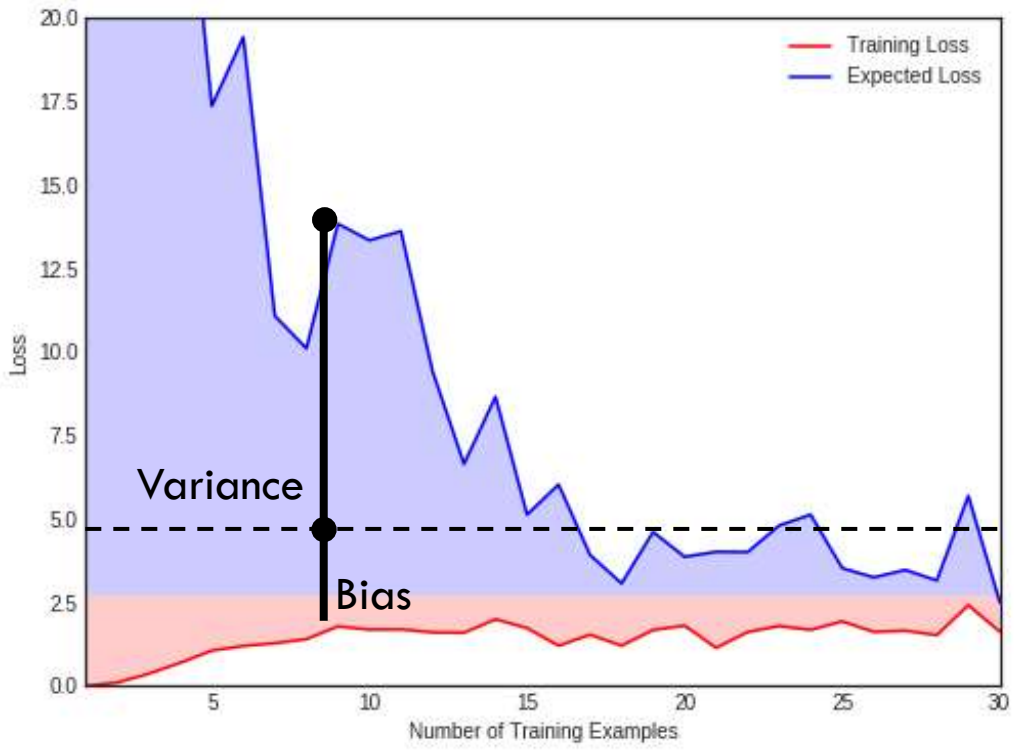


Simple Model



Complex Model

Bias–Variance on Curve



Prelude:

Curing Overfitting

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Noise causes overfitting

Overfitting is the disease

Noise is the cause

Learning is led astray by fitting the noise more than the signal

Two Cures:

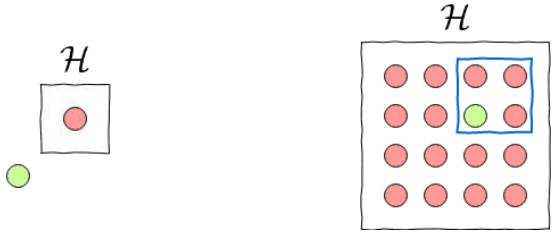
1. **Regularization**: Restrain the model
2. **Validation**: Reality check by peeking at (the bottom line)

Summary

Bias-Variance Tradeoff

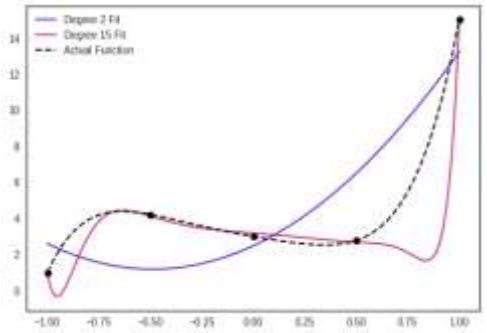
Bias: How well $\mathcal{H}$ can approximate f overall

Variance: How well we can zoom in on a good $h \in \mathcal{H}$



Match the 'model complexity' to the **data resources**, not to the target complexity

Overfitting: Fitting the data more than is warranted



Two causes: stochastic + deterministic noise

Bias $\equiv$ deterministic noise